

Meeting Summaries of November 6

Date: November 6

Time: 8:00–9:00 PM(PDT)

Attendees: Dr. Javed, Dr. Yue and Feng

Topic: Feng's report on work of sediment contamination assessment and his mistakes and lessons learned so far.

Summary of Discussion

- Feng's report on sediment contamination assessment work, it rephrases this work into the core ideas as follows:
 - The most variant factor(s) is not necessarily the most important latent pollutant factor/source that contributes to contamination assessment.
 - Human and invertebrate can hold different values of pollutants. To build a creature indicator of pollution, the focuses should not be exclusively on the latent factors that are only meaningful to humans, but also those that are meaningful to invertebrates. Leveraging the available data, translating the invertebrate's language to pollutants and **balance the views of both sides about the pollution** is the key to build a good indicator ¹.
- Feng reported the flaws and errors in his previous work - weight-driven PCA, it lacks methodological rigor and exhibits poor performance on the data, specifically:
 - The weight-driven PCs are not orthogonal in the original data space, which violates the fundamentals of PCA and weaken the meaningfulness of latent factors from it.
 - The metrics for evaluating the performance and objectiveness of the method are not well-defined; a bunch of paired t-tests and pearson correlations of eigenvectors are hard to interpret and summarize, not to mention, not to mention the flawness in its design.
 - The highly sensitive to weight changes substantially limits the searching and tuning of weights, it is a flaw in the design of the algorithm.
- Feng outlined his plan **to push the work forward to next stage: hierarchical cluster analysis on taxa groups**. The aim of this decision is to grasp a global view of the whole work, and to utilize the time and effort effectively.
- Dr. Javed suggested Feng to report the blueprint of the thesis in the next meeting, to make sure the work is on the right track and the structure is solid.

Lessons Learned So far

Since the last work on proposal (Oct 20), I spent around 14 days on designing and implementing the weight-driven PCA method that I proposed. After some proving work,

¹It is important to remind the sacrifice of accuracy in making human-based pollution assessment, since the indicator is finally serving human's purpose.

algorithms design and code implementation, it shows that the current method is flawed and not behaved as expected.

There are some lessons learned from this experience:

1. **Read and summarize more related work before start testing a "big" new idea.** Reading, summarizing the related work and comparing the published methods with the proposed method is crucial, it helps filter out the tempting prones and direct the work under a more popular framework.
2. **In my case, the best way should be to learn, summarize and reproduce the published methods.** It is great to extend the reproduced methods to my research case, and it is safer and more development-beneficial to my future work ².
3. **Coming up with new ideas is insipiring and good, but be cautious to rush into implementing them.** As the lesson 1 suggested, collecting relevant articles, summarizing and comparing with them is a more effective way to verify the new ideas. To me, the unwillingness to read and summarize is a big mistake that I have made. It is very likely due to my impetuous youth and idealistic mind towards my own brewed ideas. After several time and effort-taking trials, I realised this flaw and need to correct it right now.

Decisions Made

- A review on the whole thesis blueprint is good, checking the work is on the right track.
- Moving forward to the hierarchical cluster analysis on taxa groups, pushing the work to next stage is necessary.

Action Items

- Feng will summarize report the blueprint of the thesis in the next meeting.
- Feng will put the weight-driven PCA merthod aside, read and summarize related work about it, including the weighted-PCA, the PCA on sediment contamination assessment and the latent factor analysis on pollution source apportionment.
- Feng will tentatively use the pollution assessment results from standard PCA and pass it as input to the next step - hierarchical cluster analysis on taxa groups.

Follow-Up

Temporary meeting time: November 13, 16:00–16:45 PM (PDT)

Supportive materials

- A meeting recording with subtitles has been restored on Google Drive.

²After the algorithm design and its code implementation work, I found the work complexity is beyond my current capacity (although it worked somehow, but I am not satisfied with it). There are definitely advanced methods, both in designing or coding, that I need to learn to break this bottleneck.